

### PROFESSIONAL SUMMARY

Frontend & Full-Stack Developer specializing in modern React ecosystems, TypeScript, Python (FastAPI), and Node.js (NestJS) to engineer responsive web applications and AI-augmented SaaS platforms. Meta-certified in Front-End Engineering and UX/UI Design Principles, with a proven portfolio of 5 architected applications backed by 145 passing automated tests across five applications and CI/CD workflows. Skilled in component-driven UI architecture, client-side state management (Zustand), and secure RESTful API integrations with AI workflow orchestration.

### CORE TECHNICAL SKILLS

**Frontend Engineering:** React (18/19), TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Vite, Zustand, React Router, Framer Motion, Responsive Design, WCAG.

**AI & Intelligent Systems:** OpenAI API, Google Gemini AI, LangGraph, AI Agent Workflows, Prompt Engineering, Contextual Retrieval Patterns.

**Backend & API Systems:** Python, FastAPI, Node.js, NestJS, Express.js, RESTful APIs, PostgreSQL, Prisma ORM, Redis, Supabase (BaaS).

**DevOps, Quality & Security:** Git, GitHub Actions (CI/CD), Docker, Vitest, Jest, Unit & Integration Testing, ESLint, OWASP Security Hardening.

### FEATURED SOFTWARE ENGINEERING PROJECTS

#### CareerCraft — AI Career Platform & Multi-Agent SaaS React 19 • TypeScript • FastAPI • Python • LangGraph • Vitest • Docker

Architected a multi-service career optimization platform featuring resume ATS parsing, job matching algorithms, and automated mock interview simulations. Implemented LangGraph-driven agentic workflows in Python/FastAPI to orchestrate candidate evaluations. Enforced automated quality gates with **82 passing Vitest unit & integration tests**, trilingual i18n localization (EN, DE, AR across 258 keys), and Docker container verification.

#### Supper-Mind — Multilingual AI Wellness & Habit Analytics SaaS React • TypeScript • Tailwind CSS • Zustand • Supabase • OpenAI • Vitest

Engineered an AI wellness tracking application featuring mood analytics dashboards, habit streak calculation logic, and conversational AI coaching. Integrated Supabase authentication and PostgreSQL persistence with offline fallback mechanisms, validated across **17 automated Vitest tests**.

#### Fit-Zone — Full-Stack Fitness Management & Booking Platform React • NestJS • Prisma ORM • PostgreSQL • Redis • Gemini AI

Developed a full-stack gym management platform with NestJS, Prisma ORM, and PostgreSQL, providing dynamic class scheduling and trainer booking. Implemented secure authentication with JWT tokens, Argon2 password hashing, and role-based access control (RBAC), integrating Redis caching and Gemini AI, backed by **9 automated tests (4 frontend + 5 backend)**.

#### Mangata & Gallo — Luxury E-Commerce Platform & AI Atelier Concierge React • TypeScript • Express.js • Prisma • PostgreSQL • Tailwind CSS

Built a luxury jewelry commerce platform featuring real-time metal variant selectors, diamond carat pricing logic, and an interactive AI gift concierge. Hardened authentication session services and established HTTP security headers (nosniff, strict-origin-when-cross-origin), verified across **19 Vitest tests**.

#### Aether-Travel — Interactive Destination & Tour Discovery Platform React • TypeScript • Vite • Tailwind CSS • Leaflet Maps • Zustand

Created a travel discovery application integrating Leaflet interactive map routing, dynamic tour package filters, and centralized Zustand state management, backed by **18 integration tests**.

### CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

**Meta: Introduction to Front-End Development** (Coursera ID: 8AM5SQCXW4Z2)

**Meta: Programming with JavaScript** (Coursera ID: 2KBEK73ZYGXPX)

**Meta: Version Control (Git & GitHub)** (Coursera ID: 34D46J5FKM5W)

**Meta: Principles of UX/UI Design** (Coursera ID: V9RRQKLQDHUS)

### EDUCATION & LANGUAGES

#### Bachelor of Arts in English Language & Literature (BA)

The Higher Institute of Languages, Cairo (1998) • Grade: Good (70.80%)

**Languages:** Arabic (Native), English (Professional Proficiency)